Reproducing the ENIGMA-OCD White-Matter Microstructure × Symptom-Severity Analysis in a Single-Site OCD Sample

A diffusion-tensor imaging (DTI) reproduction study using baseline Yale–Brown Obsessive–Compulsive Scale, Second Edition (Y-BOCS-II) scores

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Data source: a single-site rTMS clinical trial for OCD — the specific study and source laboratory are withheld to protect participant confidentiality

Source study reproduced: Piras et al. (2021), Translational Psychiatry, 11, 173.

Abstract

Background. The ENIGMA-OCD Working Group reported the largest diffusion-tensor imaging (DTI) study of obsessive–compulsive disorder (OCD) to date, identifying reduced fractional anisotropy (FA) in posterior projection and association fibres (sagittal stratum, posterior thalamic radiation) in adult patients relative to controls; critically, none of these alterations scaled with symptom severity on the Yale–Brown Obsessive–Compulsive Scale (Piras et al., 2021).

Objective. To reproduce the brain–symptom-severity component of that study in an independent single-site sample of adults with OCD enrolled in a repetitive transcranial magnetic stimulation (rTMS) trial. The sample contains no healthy controls, so only the dimensional severity analysis—the analysis ENIGMA reported as null—is reproducible.

Methods. For 15 hypothesis-driven cortico–striato–thalamo–cortical (CSTC) and posterior bundles, FA (primary) and mean, radial, and axial diffusivity (secondary) were regressed on baseline Y-BOCS-II total severity adjusting for age and sex, mirroring ENIGMA's covariates. Partial correlations, Benjamini–Hochberg FDR correction, and obsession/compulsion subscale analyses were computed (N with usable DTI = 59).

Results. No DTI metric was significantly associated with symptom severity after FDR correction: 0 of 90 tests survived (minimum uncorrected $p = .115$; maximum $|\text{partial } r| = .23$). The null held for both subscales.

Conclusion. Using ENIGMA-comparable methods, the analysis replicates the central dimensional null of Piras et al. (2021): white-matter microstructure did not scale with OCD symptom severity, consistent with a trait- rather than state-marker interpretation. Range restriction, modest power, and the absence of controls mean the result constrains rather than excludes small effects.

1 Introduction

Obsessive–compulsive disorder (OCD) is a chronic and frequently disabling psychiatric condition characterised by intrusive thoughts (obsessions) and repetitive behaviours (compulsions). Influential neurobiological models implicate dysfunction of cortico–striato–thalamo–cortical (CSTC) circuits, and an extension of these models proposes that altered structural connectivity—measurable as white-matter microstructure—contributes to the disorder (Piras et al., 2021).

Diffusion-tensor imaging (DTI) characterises white-matter microstructure through the diffusion of water in brain tissue (Basser & Pierpaoli, 1996). The most widely reported scalar is fractional anisotropy (FA), an index of the directional coherence of diffusion; mean diffusivity (MD), radial diffusivity (RD), and axial diffusivity (AD) provide complementary information about overall, perpendicular, and parallel diffusion. Lower FA and higher diffusivity are commonly, though non-specifically, interpreted as reduced microstructural organisation.

Single-site DTI studies of OCD have yielded inconsistent findings, plausibly because modest samples are underpowered to detect the small effects expected in psychiatric neuroimaging (Piras et al., 2021). The ENIGMA-OCD Working Group addressed this by pooling data from 700 adult patients, 645 adult controls, 174 paediatric patients, and 144 paediatric controls across 19 sites. Adult patients showed significantly lower FA in the sagittal stratum ($d = -0.21$) and posterior thalamic radiation ($d = -0.26$); paediatric patients showed no detectable differences. Critically, the magnitude of FA reduction was not associated with symptom severity, although lower FA in the sagittal stratum was related to younger age of onset, longer illness duration, and medication status—patterns interpreted as evidence that the alterations are trait markers of the disorder rather than correlates of current symptom load (Piras et al., 2021).

1.1 Aim and scope

The present dataset comprises adults with OCD enrolled in an rTMS trial and contains no healthy-control group, so the ENIGMA case–control contrast cannot be reproduced. What can be reproduced—and what was requested—is the dimensional brain–severity analysis: the test of whether DTI metrics covary with Y-BOCS severity. This is the analysis ENIGMA reported as null.

A note on faithfulness: ENIGMA tested severity at the meta-regression (between-site) level, relating each site's average effect size to that site's mean severity. In a single-site dataset the appropriate and more direct analogue is a subject-level regression of each DTI metric on each participant's Y-BOCS-II score, adjusting for age and sex. The subject-level approach is statistically stronger but, like ENIGMA, is cross-sectional and observational.

1.2 Hypotheses

H1 (primary). Baseline Y-BOCS-II total severity is associated with FA in CSTC/posterior bundles.

H0 (the ENIGMA expectation). Consistent with Piras et al. (2021), no DTI metric is significantly associated with severity after FDR correction.

H2 (secondary). Any severity association generalises across MD, RD, AD and across obsession/compulsion subscales.

2 Methods

2.1 Sample and design

The dataset (ocd_dti_ml_dataset_plus_clinical_characterization.csv) contains adults diagnosed with OCD assessed at baseline prior to an rTMS intervention targeting either the right orbitofrontal cortex (rOFC) or the dorsomedial prefrontal cortex (DMPFC). Each participant contributes DTI tractometry and clinical characterisation. Only baseline (visit "v2") clinical scores and baseline DTI were used, matching ENIGMA's cross-sectional design.

2.2 Symptom severity

Severity was measured with the Yale–Brown Obsessive–Compulsive Scale, Second Edition (Y-BOCS-II; Storch et al., 2010), comprising obsession and compulsion subscales. The total score is the primary severity index; subscales are examined secondarily. The original ENIGMA study used the first-edition Y-BOCS (Goodman et al., 1989); both index the same construct.

2.3 DTI metrics and white-matter bundles

For every bundle the dataset provides FA, MD, RD, and AD. Following ENIGMA, FA is the primary outcome and MD/RD/AD are secondary. Analyses were restricted to a hypothesis-driven subset of bundles corresponding to the CSTC circuit and the posterior projection/association fibres ENIGMA implicated. Because this dataset uses a tractometry (pyAFQ-style) bundle atlas rather than ENIGMA's TBSS/JHU ROI atlas, bundles were mapped to the nearest ENIGMA-relevant homologues (Table M1).

Dataset bundle	ENIGMA-relevant homologue / rationale
Optic radiation (L/R)	Posterior thalamic radiation — ENIGMA's strongest case–control effect
Thalamic radiation (L/R)	Thalamo-cortical projection (CSTC)
Inferior fronto-occipital fasciculus (L/R)	Traverses the sagittal stratum — ENIGMA effect
Inferior longitudinal fasciculus (L/R)	Traverses the sagittal stratum — ENIGMA effect
Superior longitudinal fasciculus (L/R)	Fronto-parietal association fibre implicated in OCD
Cingulum (L/R)	Limbic CSTC association fibre
Corpus callosum	Major commissure; interhemispheric integration
Corticostriatal tract (L/R)	Direct structural substrate of the CSTC model

Table M1. Mapping of the 15 hypothesis-driven bundles to ENIGMA-relevant regions.

2.4 Statistical analysis

For each (metric, bundle) pair an ordinary-least-squares model was fitted: DTI metric ~ Y-BOCS-II + age + sex, mirroring ENIGMA's age + sex adjustment. The effect of interest is the Y-BOCS-II coefficient, reported as a partial correlation (rp) computed from the model t-statistic, with 95% confidence intervals via the Fisher z transform. Each analysis used all participants with complete data for that bundle (per-tract complete-case); analytic N is reported per test. Within each metric/severity family, p-values across the 15 bundles were corrected with the Benjamini–Hochberg false-discovery-rate procedure (q; Benjamini & Hochberg, 1995), with significance at $q < .05$. The minimum detectable effect at 80% power was computed to contextualise null results. Analyses used Python with pandas, NumPy, SciPy, statsmodels, and Matplotlib.

3 Results

3.1 Sample characteristics

Table 1 summarises the analytic sample. Note the restricted severity range: as an rTMS trial with a severity-based inclusion criterion, all participants fall in the moderate-to-severe band of the Y-BOCS-II (observed total range 20–40), which is relevant to interpreting the null severity association.

Characteristic	Value
N (total records)	80
N with usable DTI	59
Age, years — M (SD)	35.1 (12.3)
Age, range	18–63
Sex — Female / Male / Other, n	41 / 34 / 2
rTMS target — rOFC / DMPFC, n	38 / 38
Y-BOCS-II total (v2) — M (SD)	29.8 (4.3)
Y-BOCS-II total — observed range	20–40
Y-BOCS-II obsession (v2) — M (SD)	15.1 (2.6)
Y-BOCS-II compulsion (v2) — M (SD)	14.7 (2.3)
Age of OCD onset, years — M (SD)[1]	10.6 (5.9)
On any SRI medication, n (%)	37 (46%)
Comorbid MDD episode, n (%)	34 (42%)

Table 1. Baseline sample characteristics. [1]Onset age auto-extracted from clinical text and approximate.

Figure 1. Distribution of baseline Y-BOCS-II severity (note range restriction)

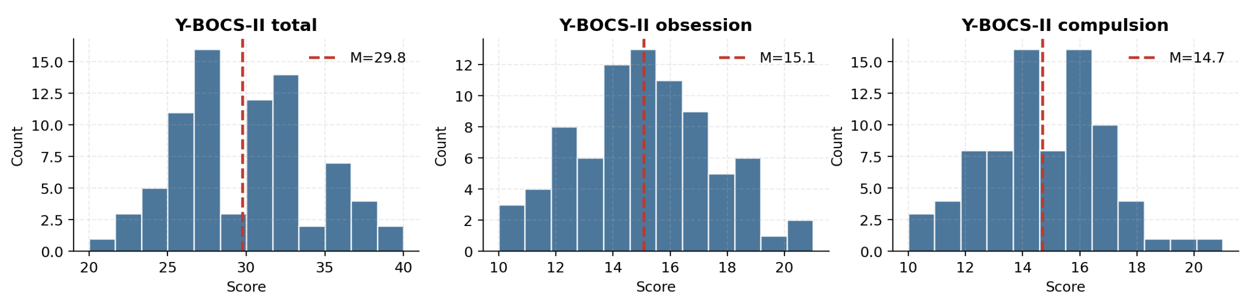

Figure 1. Distribution of baseline Y-BOCS-II severity scores. The compressed, moderate-to-severe range reflects trial inclusion criteria.

3.2 Primary analysis — FA and Y-BOCS-II total severity

Table 2 reports the primary test: partial correlations between FA and Y-BOCS-II total severity across the 15 bundles, adjusting for age and sex, with FDR correction. No bundle reached significance; all 95% confidence intervals included zero, and the strongest association (right SLF, rp = −.21) was non-significant ($p = .136$, $q = .83$).

Bundle	N	Partial r	95% CI	p	q (FDR)
Optic radiation L	49	−0.15	[−0.42, +0.15]	.332	.830

Bundle	N	Partial r	95% CI	p	q (FDR)
Optic radiation R	48	+0.04	[−0.26, +0.33]	.787	.926
Thalamic radiation L	54	−0.01	[−0.28, +0.27]	.949	.949
Thalamic radiation R	55	−0.06	[−0.32, +0.22]	.689	.926
IFOF L	52	−0.04	[−0.31, +0.25]	.802	.926
IFOF R	51	−0.19	[−0.45, +0.10]	.198	.830
ILF L	56	−0.11	[−0.37, +0.17]	.435	.837
ILF R	57	−0.14	[−0.39, +0.13]	.311	.830
SLF L	57	−0.14	[−0.39, +0.13]	.305	.830
SLF R	57	−0.21	[−0.45, +0.07]	.136	.830
Cingulum L	55	−0.10	[−0.37, +0.17]	.464	.837
Cingulum R	56	−0.19	[−0.44, +0.09]	.174	.830
Corpus callosum	57	−0.09	[−0.35, +0.18]	.502	.837
Corticostriatal tract L	56	−0.04	[−0.31, +0.23]	.762	.926
Corticostriatal tract R	56	−0.02	[−0.29, +0.25]	.872	.935

Table 2. FA versus Y-BOCS-II total severity, age- and sex-adjusted. No bundle survives FDR correction (all q > .05).

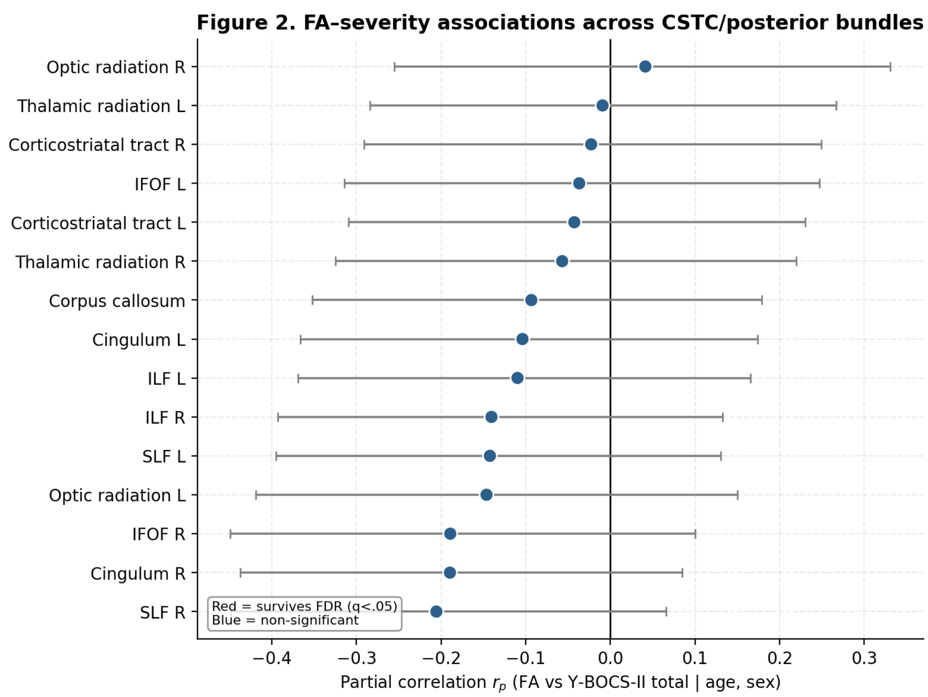

Figure 2. Forest plot of FA–severity partial correlations across the 15 bundles. All confidence intervals cross zero; none survives FDR.

3.3 Secondary metrics and subscales

Extending the total-severity analysis to MD, RD, and AD (60 tests across the four metrics) yielded no FDR-significant associations; the minimum uncorrected p was .115 and the maximum |partial r| was .23 (Figure 3). Analysing FA against the obsession and compulsion subscales separately (30 further tests) likewise produced no significant associations (minimum uncorrected p = .088). Across all 90 tests, none survived FDR correction, and no coherent cross-metric pattern (e.g., lower FA paired with higher RD) emerged.

Figure 3. DTI metric–severity partial correlations (* uncorrected p<.05; none survive FDR)

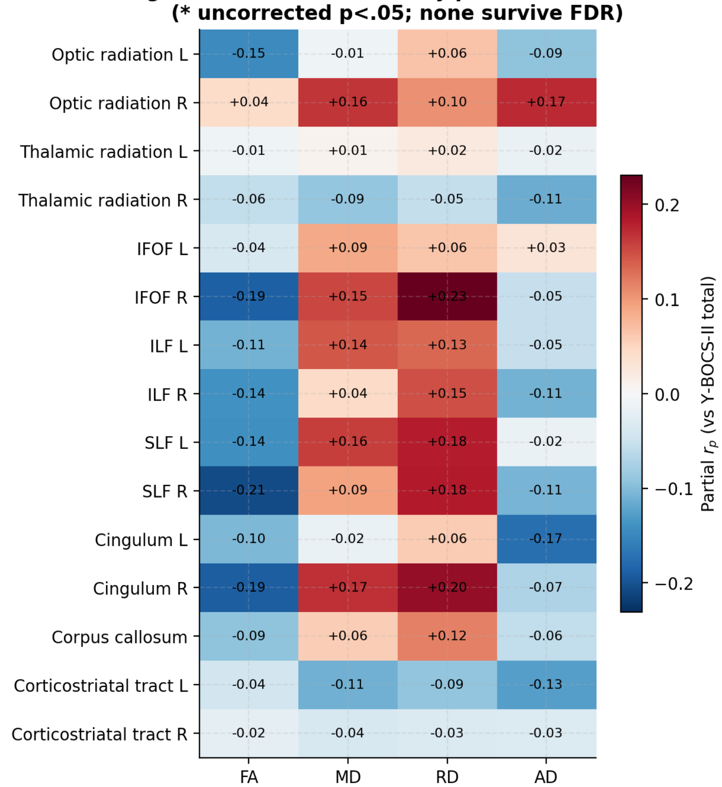

Figure 3. Partial correlations between each DTI metric and Y-BOCS-II total severity for every bundle (red = positive, blue = negative). Asterisks mark uncorrected p < .05; none survives FDR.

3.4 Strongest observed association and statistical power

Even the numerically strongest association—right SLF FA—shows no meaningful linear relationship with severity once age and sex are partialled out (Figure 4). A sensitivity analysis indicated that, at the median analytic N (≈56), the minimum partial correlation detectable with 80% power (two-tailed α = .05) was approximately r = 0.37. ENIGMA-scale severity effects (|r| < 0.1) therefore fall well below this sample's detection threshold, so the null is best read as “no moderate-or-larger severity association.”

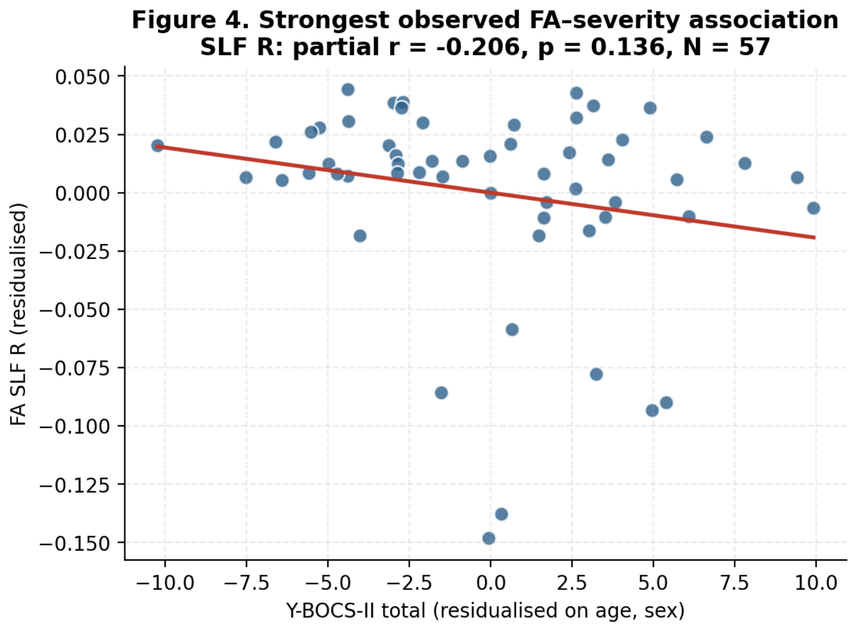

Figure 4. Added-variable plot for the strongest observed association (right SLF FA vs Y-BOCS-II total, residualised on age and sex).

3.5 Comparison with ENIGMA-OCD

Feature	ENIGMA-OCD (Piras et al., 2021)	This reproduction
Design	Cross-sectional, 19 sites (meta-analytic)	Cross-sectional, single site (subject-level)
Adult OCD N	700 (vs 645 controls)	59 (no control group)
Severity measure	Y-BOCS (Goodman et al., 1989)	Y-BOCS-II (Storch et al., 2010)
Severity test	Meta-regression of FA effect size on site severity	Subject-level OLS: DTI ~ severity + age + sex
Covariates	Age, sex (+ nonlinear, age × sex)	Age, sex
Multiple comparisons	FDR	FDR (Benjamini–Hochberg)
DTI–severity association	Not significant	Not significant (0/90 tests survive FDR)
Interpretation	Trait markers, not severity-scaled	Replicates the ENIGMA null

Table 3. Side-by-side comparison of the ENIGMA-OCD severity analysis and the present reproduction.

4 Discussion

4.1 Summary

Across 15 CSTC/posterior bundles and four DTI metrics, baseline Y-BOCS-II severity was not significantly associated with white-matter microstructure after FDR correction, and this held for obsession and compulsion subscales analysed separately. Observed associations were small and statistically indistinguishable from zero. This reproduces the central severity result of ENIGMA-OCD (Piras et al., 2021), in which white-matter alterations were unrelated to symptom severity. ENIGMA interpreted this dissociation—FA reductions related to age of onset, illness duration, and medication, but not current severity—as evidence that the microstructural signature of OCD behaves as a trait marker rather than a state marker that tracks symptom load. The present single-site, subject-level data are consistent with that interpretation.

4.2 Interpretation

A null severity association can reflect two non-exclusive possibilities: that white-matter microstructure genuinely does not scale linearly with current severity in OCD (the trait-marker account), or that the present design is not positioned to detect such scaling. The convergence with ENIGMA is notable because ENIGMA, with two orders of magnitude more participants, also failed to find a severity association—strengthening the trait-marker reading while not excluding small effects.

4.3 Limitations

1. No healthy-control group. The ENIGMA case–control contrast cannot be reproduced; this study addresses only the severity analysis—the null portion of ENIGMA.
2. Severity range restriction. Trial inclusion compressed Y-BOCS-II scores into the moderate-to-severe band (Figure 1); restricted predictor variance attenuates correlations toward null independently of any true relationship.
3. Statistical power. With median $N \approx 56$, the minimum reliably detectable partial correlation is ≈ 0.37; small ENIGMA-scale effects are undetectable here.
4. Atlas non-equivalence. Tractometry bundles were mapped to the nearest JHU/TBSS homologues, so anatomical correspondence with ENIGMA is approximate.
5. Instrument difference. Severity used Y-BOCS-II rather than the first-edition Y-BOCS; the two are highly correlated but not identical in scaling.
6. Cross-sectional, single-site, observational design precludes causal inference, and missing DTI data (handled by per-bundle complete-case analysis) modestly reduce N.

4.4 Conclusion

Using ENIGMA-comparable methods (age- and sex-adjusted regression, FDR correction, FA-primary), this reproduction found no significant association between white-matter DTI metrics and OCD symptom severity, replicating the key dimensional null reported by the ENIGMA-OCD Working Group (Piras et al., 2021). The convergence supports conceiving OCD-related white-matter microstructure as a trait-level feature rather than a marker that scales with current severity—while the sample's lack of controls, range restriction, and modest power mean the result constrains, rather than rules out, small severity effects.

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